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In [1]: #THUSHAR K P, ROLL : B24CTA60
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import plot_tree
import matplotlib.pyplot as plt
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In [2]: data=pd.read_csv("car.csv")
display(data)
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	Obj	Eg_No	Color	Type	Origin	Stolen
0	1	1	Red	Sports	Domestic	Yes
1	2	2	Red	Sports	Domestic	No
2	3	3	Red	Sports	Domestic	Yes
3	4	4	Yellow	Sports	Domestic	No
4	5	5	Yellow	Sports	Imported	Yes
5	6	6	Yellow	SUV	Imported	No
6	7	7	Yellow	SUV	Imported	Yes
7	8	8	Yellow	SUV	Domestic	No
8	9	9	Red	SUV	Imported	No
9	10	10	Red	Sports	Imported	Yes

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In [3]: le=LabelEncoder()
for column in data.columns:
    data[column]=le.fit_transform(data[column])
display(data)
```

	Obj	Eg_No	Color	Type	Origin	Stolen
0	0	0	0	1	0	1
1	1	1	0	1	0	0
2	2	2	0	1	0	1
3	3	3	1	1	0	0
4	4	4	1	1	1	1
5	5	5	1	0	1	0
6	6	6	1	0	1	1
7	7	7	1	0	0	0
8	8	8	0	0	1	0
9	9	9	0	1	1	1

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In [4]: x=data.drop(['Stolen'],axis=1)
y=data['Stolen']
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=1,random_state=42)
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In [5]: tree=DecisionTreeClassifier(criterion="entropy")
tree.fit(x_train,y_train)
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Out[5]: DecisionTreeClassifier
DecisionTreeClassifier(criterion='entropy')
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In [6]: y_pred=tree.predict(x_test)
display(y_pred)

array([1])
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In [7]: acc=accuracy_score(y_pred,y_test)
display(acc)

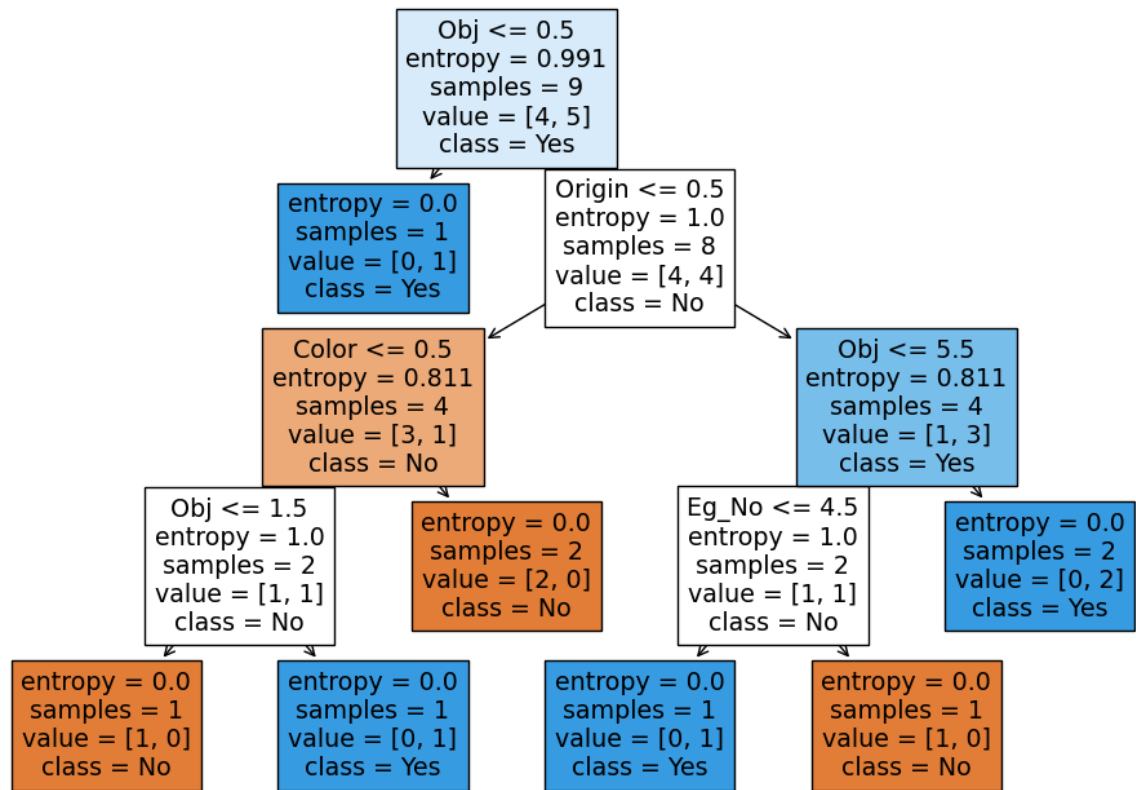
0.0
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In [8]: splittingfeat=tree.tree_.feature
display(splittingfeat)

array([ 0, -2,  4,  2,  0, -2, -2, -2,  0,  1, -2, -2, -2], dtype=int64)
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In [9]: plt.figure(figsize=(12,8))
plot_tree(tree,feature_names=list(x.columns),class_names=['No','Yes'], filled=True)
plt.show
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Out[9]: <function matplotlib.pyplot.show(close=None, block=None)>
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